Basic Quantities and Models

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1 Introduction

Survival analysis is a specialized branch of statistics designed to analyze the expected duration of time until one or more events happen, typically referred to as the **time-to-event**. Unlike standard regression models, survival analysis is uniquely equipped to handle ***censored*** datainstances where the event of interest has not occurred by the end of the study period or where subjects are lost to follow-up. Because of its versatility, this methodology is foundational across a wide spectrum of disciplines, including medicine, public health, social science, and engineering. In the biomedical sciences, researchers utilize these methods to monitor patient trajectories, quantifying the time until a specific endpoint such as death, disease relapse, or recovery. Here, the primary objective is often the estimation of survival probability over timevisualized through survival curvesor the comparison of efficacy between different treatment groups. Similarly, social scientists apply these techniques, often under the name **event history analysis**, to study the timing of life-course transitions, such as employment duration (How long in a job before leaving), marriage (At what age do people typically marry), birth intervals (How long between first and second child), or arrest recidivism (How long after release does someone get re-arrested). The field of engineering has also been a major contributor to the methodological development of survival analysis; in industrial contexts, it is frequently termed **reliability theory or failure time analysis**, focusing on modeling the lifespan of machinery and the failure rates of electronic components.

Definition: Survival time measures the duration elapsed from a specific *time origin* to the occurrence of a specific *event of interest*. Mathematically, the survival time T is defined as a non-negative random variable (i.e., $T \geq 0$). While the term stems from mortality studies, in broader applications the variable T is frequently referred to as ***failure time***, ***event time***, or ***time-to-event***.

The components of this definition can be interpreted as follows:

- **Time Scale:** The metric used to quantify the duration. This is typically measured in years, months, weeks, or days from the beginning of an individual's follow-up. Alternatively, **time** can refer to the age of an individual when the event occurs.
- **The Event (Failure):** While historically termed **failure**, this implies any designated experience of interest. Examples include death, disease incidence, relapse (return or worsening of the condition after a period of remission) from remission (A period where a condition e.g., cancer, substance use disorder, depression, is under control, not showing active symptoms/disease signs.), recovery (e.g., return to work), or component failure in engineering.

To determine survival time precisely, three fundamental requirements must be met:

1. The **time origin** must be clearly and unambiguously defined (e.g., date of diagnosis, date of surgery, or study enrollment).
2. A **time scale** for measuring the passage of time must be specified.
3. The definition of the **event** (often referred to as failure) must be precise, clear, and objective.

The specific difficulties in survival analysis arise largely from the fact that data is often incomplete. While some individuals experience the event during the observation window, others may not experience the event by the end of the study or may be lost to follow-up. Consequently, their actual survival times are unknown, providing only partial information. This phenomenon is formally known as **censoring**.

Examples of Survival Time (T):

- **Clinical Trials:** The time measured from the start of treatment (or randomization) to a specific failure event, such as disease progression or adverse reaction.
- **Demography:** The time from birth to death, which is equivalent to an individual's *age at death*.
- **Cardiology:** The time elapsed from a specific treatment intervention (e.g., bypass surgery) until cardiovascular death (That is, survival time after treatment until death due to cardiovascular causes).
- **Infectious Disease Epidemiology:** The time from the estimated moment of infection (exposure) to the onset of clinical disease symptoms (often called the *incubation period*).
- **Genetic Epidemiology:** The time from birth to the clinical manifestation of a genetic disease, commonly referred to as the *onset age*.

2 Basic Quantities and Models

Let T be a non-negative continuous random variable denoting the survival time. The distribution of survival times is fully characterized by any one of three mathematically equivalent functions: the **survival function**, the **probability density function (PDF)**, or the **hazard function**.

The Survival Function

The survival function, denoted by $S(t)$, is defined as the probability that the survival time is greater than t. It represents the probability that a subject will survive past time t. Mathematically:

$$S(t) = P(T > t) = 1 - P(T \leq t) = 1 - F(t), \quad t \geq 0,$$

where $F(t)$ is the cumulative distribution function (CDF) of T.

As t ranges over $[0, \infty)$, the survival function $S(t)$ exhibits the following fundamental properties:

- **Monotonicity:** It is a non-increasing function over time.
- **Initialization:** At time $t = 0$, $S(0) = 1$. This implies that the probability of surviving past the start time is 1.
- **Limit:** As $t \to \infty$, $S(t) \to 0$. This implies that, in the absence of censoring, every individual will eventually experience the event.

The Probability Density Function

For a continuous random variable T, the survival function can be expressed in terms of the probability density function, $f(t)$:

$$S(t) = \int_t^\infty f(u), du.$$

Conversely, the PDF can be derived from the survival function. Since $S(t) = 1 - F(t)$, it follows that:

$$f(t) = \frac{dF(t)}{dt} = -\frac{dS(t)}{dt}.$$

This establishes a one-to-one correspondence between $f(t)$ and $S(t)$.

The Hazard Function

To understand the hazard function, consider the concept of a **mortality rate**. If time is measured in discrete units (e.g., years), the mortality rate at time t is the proportion of individuals alive at time t who experience the event (fail) before time $t+1$. Mathematically, this conditional probability is written as $P(t \leq T < t + \Delta t \mid T \geq t)$.

The **hazard function**, $h(t)$, is the instantaneous limit of this rate as the time interval approaches zero. It represents the instantaneous rate at which events occur, given that the individual has survived up to time t. It is defined as:

$$h(t) = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}.$$

Using the definition of conditional probability, this relationship can be simplified to a ratio of the PDF and the survival function:

$$h(t) = \frac{f(t)}{S(t)}.$$

The Cumulative Hazard Function

The **cumulative hazard function**, $H(t)$, describes the accumulated risk up to time t. It is defined as the integral of the hazard function:

$$H(t) = \int_0^t h(u), du.$$

There is a crucial relationship linking the survival function and the cumulative hazard function:

$$S(t) = \exp\left(-H(t)\right) = \exp\left(-\int_0^t h(u), du\right).$$

This identity confirms that if we know any one of the functions $S(t)$, $f(t)$, $h(t)$, or $H(t)$, we can mathematically derive from others.

$$\begin{aligned} h(t) &= -\frac{d}{dt}(log(S(t))) \\ H(t) &= -log(S(t)) \\ S(t) &= exp(-H(t)). \end{aligned}$$

The hazard rate is **NOT** a probability, it is a **probability rate**. Therefore it is possible that a hazard rate can exceed one.

Now we will redefine 1) probability density function, $f(t)$; 2) survival function, $S(t)$; 3) hazard function, $h(t)$; and 4) cumulative hazard function, $H(t)$:

1) Survival function, $S(t)$

$$\begin{aligned} H(t) &= \int_0^t h(t)dt = \int_0^t \frac{f(t)}{S(t)}dt \\ &= \int_0^t \frac{f(t)}{1-F(t)}dt = -ln(1-F(t)) = -lnS(t) \\ S(t) &= exp\left(-H(t)\right) \end{aligned}$$

2) Probability density function, $f(t)$

$$\begin{aligned} f(t) &= \lim_{\Delta t \to 0} \frac{P(t \le T < t + \Delta t)}{\Delta t} \\ &= h(t)S(t) \\ &= h(t)exp\left(-\int_0^t h(s)ds\right) \end{aligned}$$

3) Hazard function, $h(t)$;

$$\begin{aligned} h(t) &= \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t | T \geq t)}{\Delta t} \\ &= \frac{f(t)}{S(t)} \\ &= -\frac{d}{dt}(log(S(t))) \end{aligned}$$

4) Cumulative hazard function, $H(t)$:

$$\begin{aligned} H(t) &= \int_0^t h(s)ds \\ &= \int_0^t -\frac{d}{ds}(log(S(s)))ds \\ &= -\frac{d}{ds}\int_0^t log(S(s))ds \\ &= -log(S(t)) \end{aligned}$$

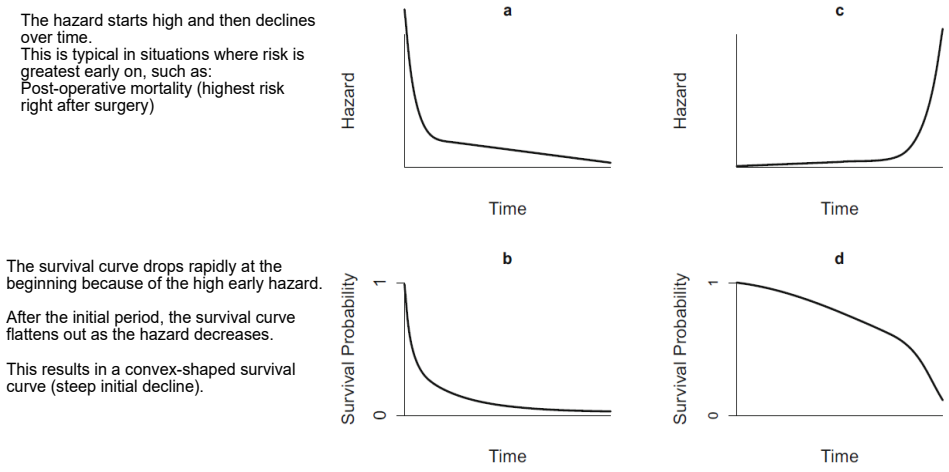


Fig. 2.1 Hazard and survival functions with high initial hazard (**a** and **b**) and low initial hazard (**c** and **d**)

Figure 1: Connection between survival and hazard functions (Source: D.F. Moore (2016))

Hazard and survival functions offer alternative representations of a survival distribution. To illustrate the relationship between the two, first consider a scenario where the hazard is initially very high, a pattern characteristic of survival times for lung cancer patients shortly after diagnosis. Figure 1(a) depicts this hazard function, while Figure 1(b) shows the corresponding survival function. Conversely, consider a case where the hazard is initially low but increases later in life, which is typical of biological aging. This pattern is illustrated in the hazard function in Figure 1(c) and the corresponding survival function in Figure 1(d).

Mean Survival time: Let $\mu = E(T)$. Due to censoring, sample mean of observed survival times is no longer an unbiased estimate of $\mu = E(T)$. If we can estimate $S(t)$ well, then we can estimate $\mu = E(T)$ using the following formula:

$$E(T) = \int_0^\infty S(t)dt.$$

3 Parametric distributions

When survival data exhibits a predictable pattern, parametric distributions can be effectively used to model failure times. A key advantage of the parametric approach is that it facilitates estimation and yields smoother survival curves by leveraging information from the entire dataset. Various parametric distributions are available for this purpose, each defined by a distinct hazard function, as detailed in Table 1.

Table 1: Hazard rates, survival functions, probability density functions, and expected lifetimes for some common parametric distributions.

Distributions	Hazard rate (h(x))	Survival function (S(x))	pdf (f(x))	Mean (E(x))
Exponential $\lambda > 0,\ x \geq 0$	λ	$exp(-\lambda x)$	$\lambda exp(-\lambda x)$	$1/\lambda$
Weibull $\alpha, \lambda > 0,\ x \geq 0$	$\alpha\lambda x^{\alpha-1}$	$exp(-\lambda x^{\alpha})$	$\alpha\lambda x^{\alpha-1} exp(-\lambda x^{\alpha})$	$\Gamma(1+1/\alpha)\lambda^{-1/\alpha}$
Gamma $\beta, \lambda > 0,\ x \geq 0$	$f(x)/S(x)$	$1 - I(\lambda x, \beta)$ $I(\lambda x, \beta) = \int_0^{\lambda x} u^{\beta-1} exp(-u) du / \Gamma(\beta)$	$\frac{\lambda^{\beta} x^{\beta-1} exp(-\lambda x)}{\Gamma(\beta)}$	β/λ
Normal $\sigma > 0,\ -\infty < x < \infty$	$f(x)/S(x)$	$1 - \Phi\left(\frac{x-\mu}{\sigma}\right)$	$\frac{1}{\sqrt{2\pi\sigma^2}} exp\left[\frac{-(x-\mu)^2}{2\sigma^2}\right]$	μ
log-normal $\sigma > 0,\ x \geq 0$	$f(x)/S(x)$	$1 - \Phi\left(\frac{logx-\mu}{\sigma}\right)$	$\frac{1}{\sqrt{2\pi x^2\sigma^2}} exp\left[\frac{-(logx-\mu)^2}{2\sigma^2}\right]$	$exp(\mu + 0.5\sigma^2)$
Extreme value $-\infty < x < \infty$	$f(x)/S(x)$	$exp(-e^Z), Z = (x-\mu)/\sigma$	$exp(Z - e^Z)$	—
Log-logistic	$\frac{\alpha x^{\alpha-1}\lambda}{1+\lambda x^{\alpha}}$	$\frac{1}{1+\lambda x^{\alpha}}$	$\frac{\alpha x^{\alpha-1}\lambda}{(1_\lambda x^{\alpha})^2}$	$\frac{\pi Csc(\pi/\alpha)}{\alpha\lambda^{1/\alpha}}$

Here, λ is the scale parameter, and α is the shape parameter for the exponential and Weibull distributions. Additionally, β is the shape parameter for the gamma distribution

To analyze the data, we usually plot the observed hazard function to see if it aligns with a specific distribution. If the data supports a parametric assumption, these methods are generally preferred over non-parametric ones for identifying risk factors. The most important parametric forms in medical science are the exponential, Weibull, and log-normal distributions.

3.1 The exponential distribution

The exponential distribution is the simplest parametric model used in survival analysis, defined uniquely by a single rate parameter, λ. The defining feature of this distribution is that its hazard rate remains constant over time (as illustrated in Figure 2). This implies that the risk of failure is purely random and does not depend on how long the subject has already survived.

This characteristic leads to the **memoryless property**, which posits that an object does not age or experience "wear and tear." Mathematically, this is expressed as $P(T > s + t \mid T > s) = P(T > t)$. For example, consider a light bulb with an exponentially distributed lifespan. The probability that it fails in the next instant is constant, regardless of whether the bulb is new or has been burning for years. To visualize this empirically, one can plot the logarithm of the cumulative hazard against the logarithm of time. If the data follows an exponential distribution, this plot will yield a straight line with a slope of 1 (a 45-degree angle).

Example of the **Memoryless Property**: For a component with an exponentially distributed lifetime, the probability that a one-year-old component lasts 3 more months in operation is identical to the probability that a ten-year-old component lasts 3 more months. The component effectively has no "memory" of its age.

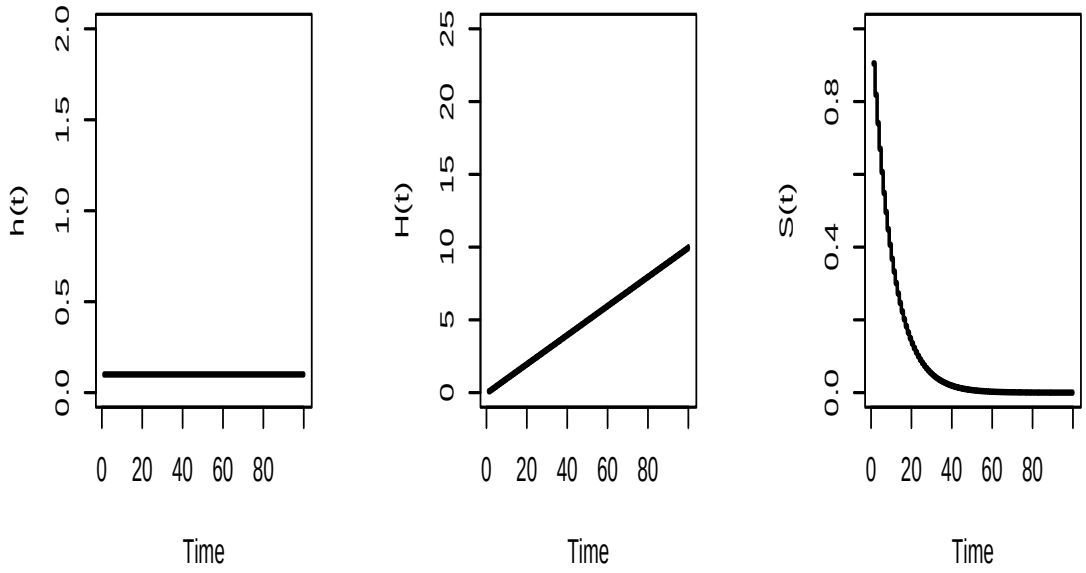


Figure 2: Hazard rate, cumulative hazard and survival as a function of time for the exponential distribution when $\lambda = 0.25$.

3.2 The Weibull distribution

The Weibull distribution is described by a scale parameter λ and shape parameter α. If $\alpha < 1$, hazard rate monotonically decreases with time, if $\alpha = 1$, hazard rate is constant over time (equivalent to the exponential distribution) and if α is greater than 1 hazard rate increases with time. Figure 3 is an example of a Weibull distributed survival pattern with $\alpha < 1$. Failure time data can be checked for consistency with the Weibull distribution by plotting the log cumulative hazard as a function of log time: Weibull distributions in this case will yield a straight line.

The Weibull distribution is a more flexible generalization of the exponential distribution, described by a scale parameter λ and a shape parameter α. The value of α determines the behavior of the hazard function, allowing the model to adapt to different types of survival data:

- $\alpha < 1$ (Decreasing Hazard): The hazard rate decreases monotonically over time. This models "infant mortality" or early failure, where items are most likely to fail immediately after manufacturing or diagnosis (see Figure 3).

- $\alpha = 1$ (Constant Hazard): The hazard rate is constant. In this specific case, the Weibull distribution reduces to the exponential distribution.
- $\alpha > 1$ (Increasing Hazard): The hazard rate increases over time. This models the process of aging or "wear-out," where the risk of death or failure grows as time passes.

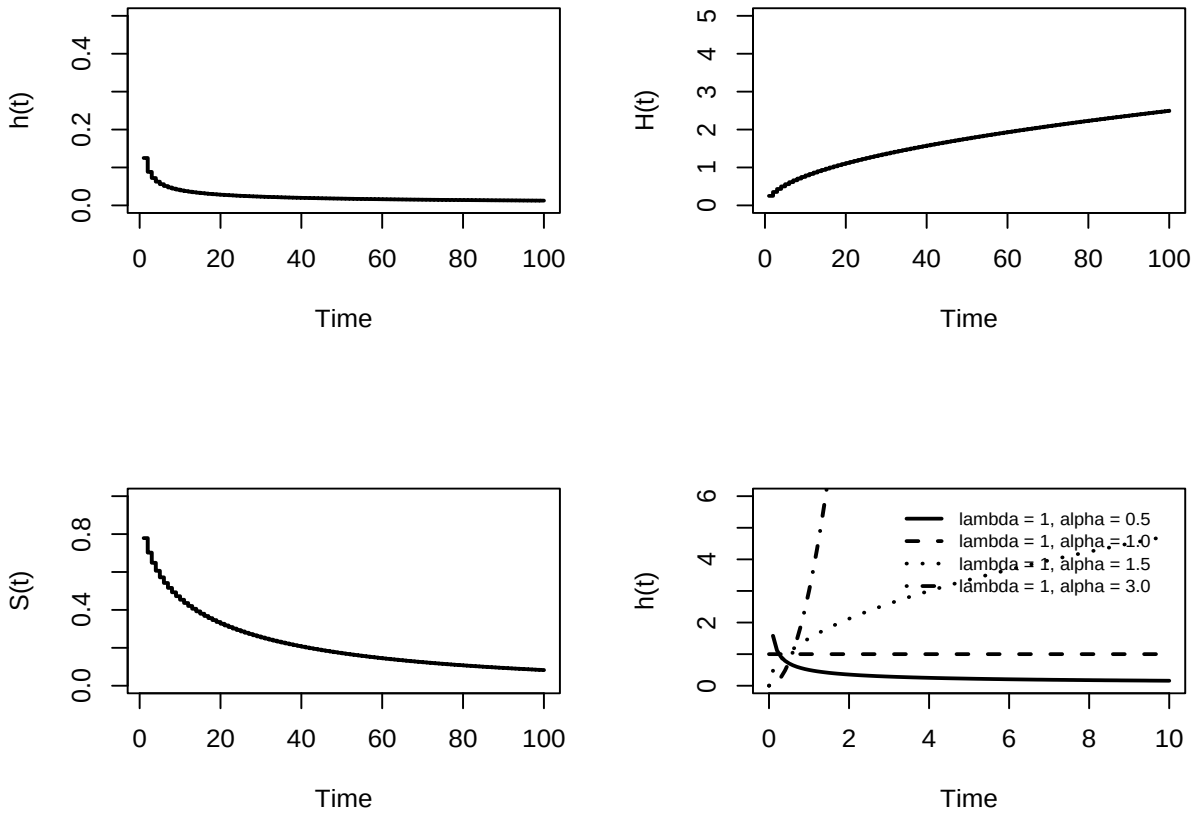


Figure 3: First three plots show that the Hazard rate, cumulative hazard and survival as a function of time for the Weibull distribution when $\lambda = 0.25$ and $\alpha = 0.5$.

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3.3 Regression Models for Survival Data

Thus far, we have modeled the survival experience of homogeneous populations. However, practical survival analysis frequently requires adjusting the survival function to account for heterogeneity among subjects. for example, cancer patients in the study are not identical. They differ in many ways that can dramatically affect their survival time. This variation is called heterogeneity. Examples of Differences (Heterogeneity Factors):

- Age: A 30-year-old vs. an 80-year-old
- Disease Severity: Stage 1 vs. Stage 4 cancer
- Treatment Received: New drug vs. standard therapy
- Co-morbidities: Patient with no other illnesses vs. patient with heart disease
- Lifestyle Factors: Smoker vs. non-smoker.

This concomitant information is captured through covariates (also referred to as explanatory or predictors or covariates). Accounting for such heterogeneity is crucial across various study designs, including clinical trials, cohort studies, and observational studies.

Consider a failure time $T > 0$ associated with a vector of covariates $\boldsymbol{x} = (x_1, x_2, \cdots, x_p)$. These covariates may include quantitative variables (e.g., blood pressure, age, weight) and qualitative variables (e.g., gender, disease Severity, treatment, co-morbidities, disease status, lifestyle factors). The objective is to model the survival function, hazard function, or other parameters conditional on these covariates. By doing so, we can compare survival across treatment groups, predict patient prognosis based on specific characteristics, and identify risk factors while controlling for confounders (variables associated with both the exposure and the failure time).

Two primary approaches for modeling covariate effects exist in the statistical literature. The first is analogous to classical linear regression. In this approach, the natural logarithm of the survival time, $y = \ln(T)$, is modeled directly. This log-transformation maps positive survival times onto the entire real line, making linear modeling appropriate. We assume the following linear model for y:

$$y = \mu + \boldsymbol{x}\boldsymbol{\beta} + \sigma W, \tag{1}$$

where $\boldsymbol{\beta} = (\beta_1, \beta_2, \cdots, \beta_p)^\top$ is a vector of regression coefficients and W is the error distribution. Common choices for the error distribution include the standard normal distribution (which yields a log-normal regression model) and the extreme value distribution (which yields an exponential or Weibull regression model).

Model (1) is known as the Accelerated Failure Time (AFT) model. To understand this terminology, let $S_0(t)$ denote the baseline survival function of T (i.e., when $\boldsymbol{x} = \mathbf{0}$); this corresponds to the survival function of $\exp(\mu + \sigma W)$. We can derive the conditional survival function as follows:

$$\begin{aligned} P(T > t|\boldsymbol{x}) &= P(Y > logt|\boldsymbol{x}) \\ &= P(\mu + \sigma W > logt - \boldsymbol{x}\boldsymbol{\beta}|\boldsymbol{x}) \\ &= P(\mu + \sigma W > log(texp(-\boldsymbol{x}\boldsymbol{\beta}))|\boldsymbol{x}) \\ &= P(e^{\mu+\sigma W} > texp(-\boldsymbol{x}\boldsymbol{\beta})|\boldsymbol{x}) \\ &= S_0(texp(-\boldsymbol{x}\boldsymbol{\beta})) \end{aligned}$$

Observe that the effect of the covariates is to rescale time by a factor of $\exp(-\boldsymbol{x}\boldsymbol{\beta})$. In other words, the AFT model states that the survival probability of an individual with covariates $\boldsymbol{x}$ at time t is identical to the baseline survival probability at time $t\exp(-\boldsymbol{x}\boldsymbol{\beta})$. Depending on the sign of $\boldsymbol{x}\boldsymbol{\beta}$, the passage of time is either accelerated or decelerated relative to the baseline. Consequently, the hazard rate for an individual with covariates $\boldsymbol{x}$ is related to the baseline hazard rate $h_0(t)$ by:

$$h(t|\boldsymbol{x}) = h_0(t\exp(-\boldsymbol{x}\boldsymbol{\beta}))\exp(-\boldsymbol{x}\boldsymbol{\beta}).$$

4 Censoring

A key characteristic of survival data is that the response variable is a non-negative discrete or continuous random variable, and represents the time from a well-defined origin to a well-defined event. A second characteristic of survival analysis is censoring which occurs the starting or end of the event is not precisely observed. That is, the individual has not yet experienced the relevant outcome, such as relapse or death, by the close of the study; the study participant is lost to follow-up during the study period; or, the individual experiences an event that prevents further follow-up. The most common example of this is right censoring, which results when the final endpoint is only known to exceed a particular value.

Censored Data: Censored data occurs when we have some information about individual's survival time, but we dont know the survival time exactly. The censoring occurs for the following situations:

- A person does not experience the event before the study ends.
- A person is lost to follow-up during the study period.
- A person withdraws from the study because of death (if death is not the event of interest) or some other reason (e.g., adverse drug reaction)

There are three types of censoring, right, left, and interval.

Right censored: If the events occur beyond the end of the study, then the data is right-censored. The person's true survival time becomes incomplete when the study ends or he is lost to follow-up or withdrawn at the right side of the follow-up period. We generally refer to this kind of data as right-censored.

Left Censored: Left-Censoring happens when we know the event of interest (e.g., disease onset, failure of a machine) occurred before the study started or before we began observing that specific individual, but we do not know exactly when it happened. In this scenario, the subject's true survival time is less than or equal to the observed survival time.

How Data Becomes Left-Censored:

- A person walks into a clinic for a routine check-up.
- They take an HIV test, and it comes back positive.
- Crucially, the person does not know (or cannot recall) the exact date of exposure. They might have been exposed weeks, months, or years ago.
- For our study, the **observed time** is the date of the positive test.

For example, if we are following subjects until they test positive for HIV, we may record a "failure" when a subject first tests positive. However, if we do not know the exact date of exposure, we only know the event occurred sometime before the test. Thus, the survival time is censored on the left side.

Another example involves a student measuring minimum daily temperatures using a thermometer with a lower limit of $-10°\text{F}$. If the actual temperature drops below $-10°\text{F}$, the thermometer will simply read $-10°\text{F}$. The resulting data point is considered left-censored because the true value is known only to be less than or equal to the threshold.

Interval Censored: Data is interval censored when a subject's true (but unobserved) survival time is known only to fall within a specific time interval. For instance, in HIV surveillance, a subject may test negative at time t_1 and subsequently test positive at time t_2. In this case, the true event time occurred somewhere between t_1 and t_2. The subject is therefore interval censored within the interval (t_1, t_2).

Left Truncation: Left truncation occurs when subjects have been at risk for the event before entering the study. This often arises when information, such as the time of diagnosis, is gathered retrospectively. For example, in a study of disease mortality where the outcome is survival from the time of diagnosis, patients may not be enrolled until months or years after their initial diagnosis. These patients represent a biased sample because they had to survive long enough to reach the enrollment date; those who died between diagnosis and enrollment are excluded. To leave them in the risk set without adjustment would bias the survival estimates, so they must be treated as left-truncated.

In practice, we most frequently deal with right censoring, though left truncation is also common in observational studies.

Median Survival time: If T is a continuous random variable, then the pth quantile is found by solving the equation $S(t_p) = 1 - p$. The median survival time is the 50th percentile $t_{0.50}$ of the distribution of T. That is, the time point where 50 percent of the subjects are still alive. The median survival time for a continuous random variable T is the value $t_{0.50}$ so that $S(t_{0.50}) = 0.50$.

Example 1: The lifetime of light bulbs follows an exponential distribution with a hazard rate of 0.001 failures per hour of use.
a) Find the mean lifetime of a randomly selected light bulb.
b) Find the median lifetime of a randomly selected light bulb.
c) What is the probability a light bulb will still function after $2,000$ hours of use?

Example 2: The time to death (in days) following a kidney transplant follows a log logistic distribution with $\alpha = 1.5$ and $\lambda = 0.01$.
a) Find the 50 and 150 day survival probabilities for kidney transplantation in patients.
b) Find the median time to death following a kidney transplant.
c) Show that the hazard rate is initially increasing and, then, decreasing over time. Find the time at which the hazard rate changes from increasing to decreasing.

5 Survival data

For subject $i = 1, 2, \cdots, n$, let T_i be the true survival time and C_i the censoring time. Then the observed event time can be defined as $Y_i = \min(T_i, C_i)$. The event indicator $\delta_i = I(T_i \leq C_i)$ takes on a value of 1 if subject i experiences the event, and 0 if the subject is censored. The observed survival data is $\{(Y_i, \delta_i)$ for $i = 1, 2, \cdots, n\}$.

The survival time data for the eight individual in the Figure 4 is presented. For each individual, we have given the corresponding survival time up to the event's occurrence or up to censorship. Our follow up time is 100 weeks. We have indicated in the end of study line whether this time was censored or not (with 1 denoting failed and 0 denoting censored). For example, Data for the second person shows a survival time of 70 weeks and a failure time indicator of 1, while the third person has a survival time of 75 weeks, is withdrawn at 75 weeks, and also has a censorship indicator of 0.

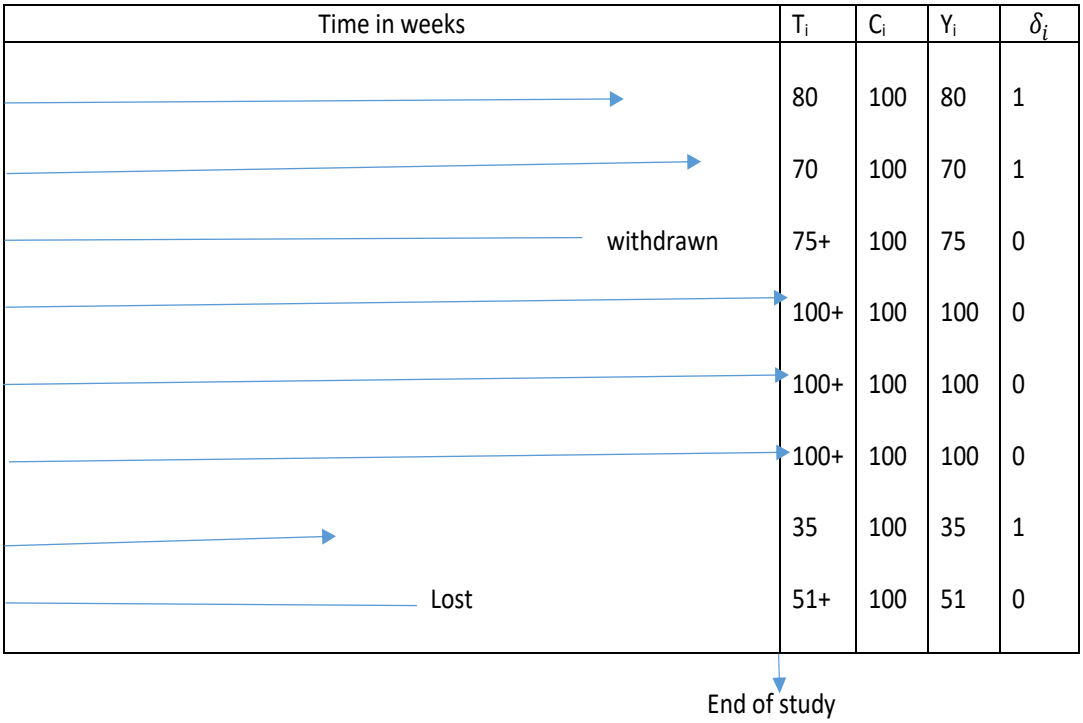


Figure 4: Hypothetical example of right censored data

When right censoring occurs, the only information known to the researcher is survival time at the occurrence of censoring. Statisticians utilize this partial information of right censoring when developing a survival model. Specifically, the information of right censored survival times can be well integrated in a likelihood function of survival data. Our potential data: $\{(T_i, C_i) \text{ for } i = 1, 2, \cdots, n\}$ and observed data $\{(Y_i, \delta_i) \text{ for } i = 1, 2, \cdots, n\}$.

Of course we are interested in making inference on the random variable T. We define the following corresponding functions for the survival time T.

- $f_T(t)$ = density function
- $F_T(t)$ = distribution function
- $S_T(t)$ = survival function
- $h_T(t)$ = Hazard function

Since we work on the observed data $\{(Y_i, \delta_i) \text{ for } i = 1, 2, \cdots, n\}$, we define the following corresponding functions for the censoring time C:

- $f_C(t)$ = density function
- $F_C(t)$ = distribution function = $P(C \leq t)$
- $S_C(t)$ = survival function = $P(C \geq t)$ = 1- $P(C \leq t) = 1 - F_C(t)$
- $h_C(t)$ = Hazard function = $\frac{f_C(t)}{S_C(t)}$

In order to derive the density of (Y, δ) (suppress i), we assume independent censoring, variables T and C are independent. The density function of (Y, δ) is defined as

$$f(y, \delta) = \lim_{h \to 0} \frac{P(t \leq Y \leq t + h, \delta = \delta)}{h}, \quad t \geq 0, \quad \delta \in \{0, 1\}$$

Case 1: $\delta = 1$, i.e., $T \leq C$, $Y = min(T, C) = T$, we have

$$\begin{aligned}
& P(t \leq Y < t + h, \delta = 1) \\
= \; & P(t \leq T < t + h, C \geq T) \\
\approx \; & P(t \leq T < t + h, C \geq y), \quad \text{(y is a fixed number)} \\
= \; & P(t \leq T < t + h) * P(C \geq y), \quad \text{independence} \\
= \; & f_T(\xi)h * S_C(y), \quad \xi \in (t, t + h)
\end{aligned}$$

Therefore

$$\begin{aligned} f(y, \delta = 1) &= \lim_{h \to 0} \frac{P(t \leq Y < t + h, \delta = 1)}{h} \\ &= \lim_{h \to 0} \frac{f_T(\xi)h * S_C(y)}{h} \\ &= f_T(y) * S_C(y), \quad \xi \in (t, t + h) \end{aligned}$$

Case 2: $\delta = 0$, i.e., $T \geq C$, $Y = min(T, C) = C$, we have

$$\begin{aligned} & P(t \leq Y < t + h, \delta = 0) \\ = \; & P(t \leq C < t + h, C \leq T) \\ \approx \; & P(t \leq C < t + h, T \leq y), \quad \text{(y is a fixed number)} \\ = \; & P(t \leq C < t + h) * P(T \geq y), \quad \text{independence} \\ = \; & f_C(\xi)h * S_T(y), \quad \xi \in (t, t + h) \end{aligned}$$

Therefore

$$\begin{aligned} f(y, \delta = 0) &= \lim_{h \to 0} \frac{P(t \leq Y < t + h, \delta = 0)}{h} \\ &= \lim_{h \to 0} \frac{f_C(\xi)h * S_T(y)}{h} \\ &= f_C(y) * S_T(y), \quad \xi \in (t, t + h) \end{aligned}$$

Combining these two cases, we have the density function of (Y, δ)

$$\begin{aligned} f(y, \delta) &= [f_T(y)S_C(y)]^{\delta}[f_C(y)S_T(y)]^{1-\delta} \\ &= [f_T(y)]^{\delta} S_T(y)]^{1-\delta}[S_C(y)]^{\delta}[f_C(y)]^{1-\delta} \\ &= [h_T(y) * S_T(y)]^{\delta} S_T(y)]^{1-\delta}[S_C(y)]^{\delta}[f_C(y)]^{1-\delta} \\ &= [h_T(y)]^{\delta} * S_T(y) * (constant), \end{aligned}$$

where the hazard function $h_T(y) = \frac{f_T(y)}{S_T(y)}$ and $h_T(y) * S_T(y) = f_T(y)$.

Example based on right censored data: Suppose the underlying survival time T is from an exponential distribution with parameter λ and we have observed data: $(Y_i, \delta_i), i = 1, 2, \cdots, n$. Since $\lambda(t) = \lambda$ and $S(t) = e^{-\lambda t}$, we get the likelihood function of λ:

$$\begin{aligned} L(\lambda; y, \delta) &= \prod_{i=1}^{n} \lambda^{\delta_i} e^{-\lambda y_i} \\ &= \lambda^{\sum_{i=1}^{n} \delta_i} e^{-\lambda \sum_{i=1}^{n} y_i} \\ \ell(\lambda; y, \delta) &= logL = \lambda \sum_{i=1}^{n} \delta_i - \lambda \sum_{i=1}^{n} y_i \end{aligned}$$

The first derivative of log likelihood,

$$\frac{d\ell}{d\lambda} = \frac{\sum_{i=1}^{n} \delta_i}{\lambda} - \sum_{i=1}^{n} y_i = 0$$

The maximum likelihood of λ is

$$\hat{\lambda} = \frac{\sum_{i=1}^{n} \delta_i}{\sum_{i=1}^{n} y_i} = \frac{\text{Number of failures}}{\text{Individual time at risk}}.$$

The second derivative of log likelihood,

$$\frac{d^2\ell}{d\lambda^2} = -\frac{\sum_{i=1}^{n} \delta_i}{\lambda^2}.$$

So the estimated variance for $\hat{\lambda}$ is

$$var(\hat{\lambda}) = -E\left(\frac{d^2\ell}{d\lambda^2}\right)^{-1} = \left[\frac{\sum_{i=1}^{n} \delta_i}{\lambda^2}\right]_{\lambda=\hat{\lambda}}^{-1} = \frac{\hat{\lambda}^2}{\sum_{i=1}^{n} \delta_i}.$$

The asymptotic distribution of $\hat{\lambda}$ is

$$\hat{\lambda} \sim N\left(\lambda, \frac{\hat{\lambda}^2}{\sum_{i=1}^{n} \delta_i}\right)$$

This result can be used to construct confidence interval or perform hypothesis testing on λ. For example, a $(1-\alpha)100\%$ confidence interval for λ is given by

$$\hat{\lambda} + Z_{\alpha/2} \frac{\hat{\lambda}}{\sqrt{\sum_{i=1}^{n} \delta_i}}.$$

Real Data Example 1: The data below show survival times (in months) of patients with certain disease

$$3, 5, 6^*, 8, 10^*, 11^*, 15, 20^*, 22, 23, 27^*, 29, 32, 35, 40, 26, 28, 33^*, 21, 24^*,$$

where * indicates right censored data. Fit the exponential model for this data set.

Answer: If we fit exponential model to this data, we have $\sum_{i=1}^{20} \delta_i = 13$, Total survival time$= \sum_{i=1}^{10} y_i = 418$. Now the estimate of λ and standard error of $\hat{\lambda}$ are

$$\hat{\lambda} = 13/418 = 0.0311/month \quad SE(\hat{\lambda}) = \hat{\lambda}/13 = 0.0086.$$

A 95% CI for λ is

$$\hat{\lambda} \pm z_{0.025} SE(\hat{\lambda}) = 0.0311 \pm 1.96 * 0.0086 = [0.0142, 0.048].$$

To see how well the exponential model fits the data, the fitted exponential survival function is superimposed to the Kaplan-Meier estimate as shown in Figure 5:

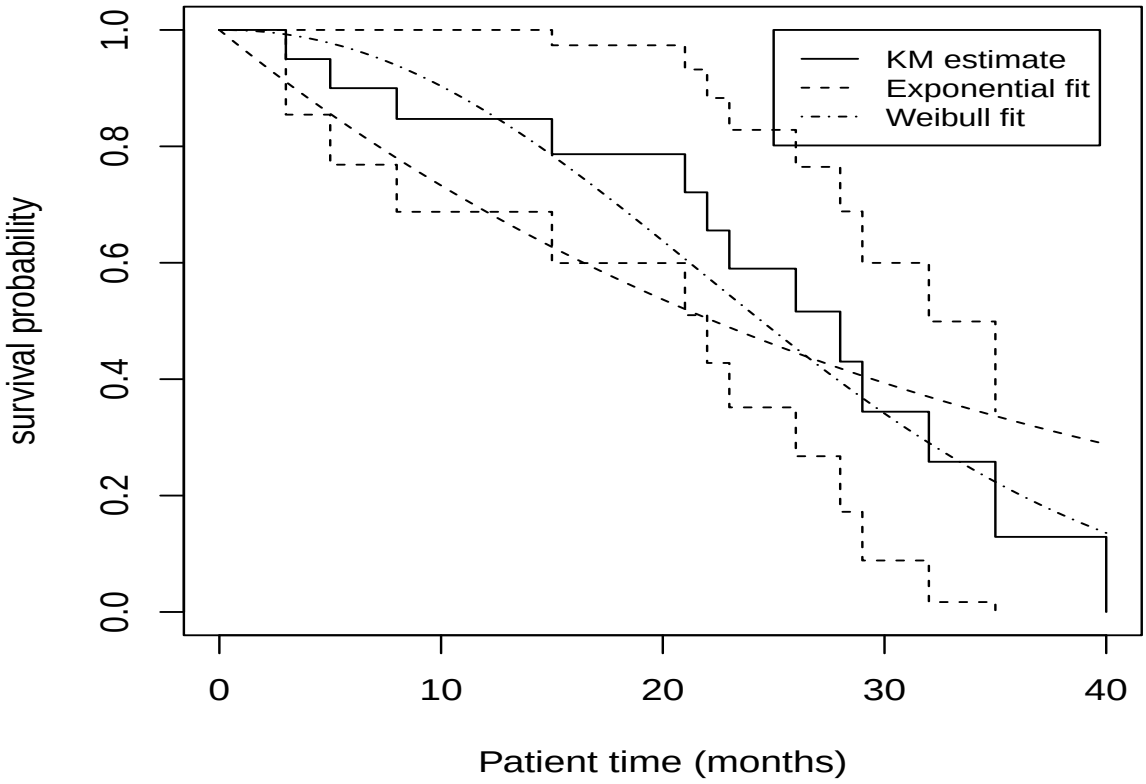


Figure 5: KM curve, exponential and Weibull fits to the survival data

.

```
rm(list=ls())
library(survival)
time=c(3, 5, 6, 8, 10, 11, 15, 20, 22, 23, 27, 29, 32, 35,
40, 26, 28, 33, 21, 24)
status=c(1,1,0,1,0,0,1,0,1,1,0,1,1,1,1,1,1,0,1,0)

# Fit KM curve, exponential
fit <- survreg(Surv(time, status)~1, dist="exponential")
summary(fit)
lambda= 1/exp(fit$coef) # scale estimate
stderror= sqrt(lambda^2/sum(status)) # standard error of lambda-hat;

# Fit Weibull model
> fit1 <- survreg(Surv(time, status)~1, dist="weibull")
> summary(fit1)

Call:
survreg(formula = Surv(time, status) ~ 1, dist = "weibull")
             Value Std. Error     z      p
(Intercept)  3.367      0.129 26.07 <2e-16
Log(scale)  -0.765      0.234 -3.27 0.0011

Scale= 0.465

Weibull distribution
Loglik(model)= -54.1   Loglik(intercept only)= -54.1
Number of Newton-Raphson Iterations: 7
n= 20

> lambda1= exp(-(fit1$icoef[1])/exp(fit1$icoef[2])) # scale estimate
> lambda1
 (Intercept)
0.0007192432
> shape1 = 1/exp(fit1$icoef[2]) # shape estimate.
> shape1
Log(scale)
  2.149375
```

```
#To see how well the exponential model fit the data

fit <- survfit(Surv(time, status)~1, conf.type=c("plain"))
plot(0,0, xlim=c(0,40), ylim=c(0,1), xlab="Patient time (months)",
ylab="survival probability", pch=" ")
lines(fit, lty=1)
x <- seq(0,40, by=0.5)
sxE <- exp(-0.0311*x)
lines(x, sxE, lty=2)

sxW = exp(-lambda1*x^shape1)
lines(x, sxW, lty=4)
legend(25,1, c("KM estimate", "Exponential fit", "Weibull fit"),
lty=c(1,2,4), cex=0.8)
```

Real Data Example 2: The data below show the remission times (in weeks) of leukemia patients undergoing a new therapy:

$$7, 10, 14^*, 18, 22^*, 25, 29, 35^*, 40, 16, 45^*, 50, 55^*, 60, 21^*$$

where * indicates right censored data. Using R, fit the Weibull model for this data set by estimating the shape (α) and scale (λ) parameters.